Campus Map Editor Manual

A complete guide to the Campus Map Editor (`index.html` + `editor.js`) and how its output is used by the public map page (`map.php`).

1. What the editor does

The editor lets you draw the campus: **buildings**, the **rooms** and **stairs** inside them, **junctions** (the walking points used for pathfinding), the **connections** between them, and plain decorative **lines** such as walls. The result is exported as a JSON layout that `map.php` loads (`map.json` , or `map-layout.json` if `map.json` does not exist).

2. Screen layout

- **Left sidebar** has three panels: Buildings, Quick actions and Properties.
- **Top toolbar** holds the tool buttons, duplicate, export, import, undo, redo and the zoom controls.
- **Canvas** is the drawing area. Its size is set in Quick actions (default 1920 x 1080).
- **Hide the bars:** the double-arrow button in the sidebar header hides the sidebar, and the eye button next to the zoom controls hides the top toolbar. Move the mouse to the screen edge and a round button appears to bring them back.
- **Status pill** (top right of the sidebar) shows the current mode and short messages such as why a connection was refused.

3. Tools and shortcuts

Tool	Key	What it does
Select	S	Click to select, drag to move, drag handles to resize.
Add building	B	Creates a building (named from the building name field if filled).
Add room	R	Click the canvas to place a room.

Tool	Key	What it does
Add junction	J	Click the canvas to place a junction.
Add stair	F	Click the canvas to place a stair.
Add line	L	Click the start point, then the end point.
Connect	C	Click two nodes to connect them (click them again to remove the connection).
Delete	Delete / Backspace	Deletes the selection.
Duplicate	Ctrl+J	Copies the selected items slightly offset.
Export JSON	button	Downloads the layout as a `.json` file.
Import JSON	button	Loads a layout file.
Undo / Redo	Ctrl+Z / Ctrl+Y (or Ctrl+Shift+Z)	Steps through the last 20 changes.

Other shortcuts:

- **Arrow keys** nudge the selected item by 10 px.
- **Ctrl+A** selects everything. Pressing **Ctrl+J, Ctrl+R, Ctrl+B or Ctrl+S** within about 0.7 seconds afterwards selects only all junctions, rooms, buildings or stairs.
- **Ctrl+C / Ctrl+X / Ctrl+V** copy, cut and paste.
- **Esc** closes the multi-edit popup.
- Shortcuts are ignored while you are typing in a text field.

4. Moving around the canvas

- **Zoom:** mouse wheel (50% to 210%), the + and - buttons, or type a percentage in the zoom box and press Enter.
- **Pan:** in Select or Connect mode, drag empty canvas. Shift or Alt plus drag, or the middle mouse button, also pans.
- **Box select:** hold the right mouse button and drag a rectangle.
- **Multi-select:** Ctrl, Cmd or Shift plus click adds or removes an item from the selection.

- **Multi-edit popup:** right-click on any item of a multi-selection to change a shared property (colour, font colour, building, floor and similar) for all of them at once.

5. Buildings

A building is a dashed rectangle that groups rooms, stairs and junctions.

- **Add:** use Add building (or press B). Each building appears in the **Buildings panel**.
- **Buildings panel:** each row has a colour swatch, an editable name, floor buttons (shown only when the building has more than one floor) and a **trash icon** to delete the building.
- **Floors:** select a building and set **Floors** in Properties, then press Apply. If you lower the number, any item that sat above the new top floor is moved down to the new top floor (you can undo this).
- **Moving a building** moves everything assigned to it. Items assigned to a building are kept inside its rectangle.
- **Deleting a building** leaves its items in place but unassigned.

6. Floors

- **Active floor** (Quick actions) is the floor shown on the canvas. The floor buttons in a building row switch just that building.
- Items are only shown on their own floor. The item you have selected is always visible.
- **Floor limit:** an item's **Floor** field only offers floors that exist in its building. A building with 2 floors offers 1 and 2. Items with no building (junctions only) can use any floor up to the tallest building.
- If you move an item into another building, its floor is reduced if the new building is shorter.

7. Rooms, stairs and junctions

- **Room:** a destination people can pick. Needs a building.
- **Stair:** links floors. Connect stairs on neighbouring floors to each other, or to junctions on the same or an adjacent floor.
- **Junction:** an invisible walking point on the public map. Rooms connect to junctions, and junctions connect to stairs. A junction may be assigned to **No building** in the Building dropdown. It then belongs to no building but still has a Floor, and on the public

map it appears only while that floor is being viewed.

8. Connections (pathfinding)

Use the Connect tool. The rules are:

- A room connects to a junction only (never directly to another room).
- A room can connect to one junction. Turn on **Multi junction connection** for that room to allow more than one.
- Junction to junction is allowed when their floors differ by at most 1.
- Stair to stair is allowed only between adjacent floors. Stair to junction is allowed on the same or an adjacent floor.
- Clicking two already-connected nodes removes the connection.
- After joining two junctions, the second junction stays ready so you can keep chaining.
- **Hide junctions & connections** (Quick actions) hides them while you work on the layout. They stay in the data.

9. Free lines

Use Add line (L): click once for the start and again for the end. Lines are always **solid black**. Select a line to drag it or drag the two end handles. Set thickness in **Line width** and angle in **Rotation**. With **Snap to grid** on, line ends snap to the grid and to other lines' ends. Lines are decoration only and are not part of pathfinding.

10. The Properties panel

Select an item to edit it. **Changes are only committed when you press Apply or press Enter in a field.** If you click a different item first, anything you typed but did not apply is discarded and is never applied to the other item. Checkboxes and the Building dropdown take effect immediately on the currently selected item.

Field	Applies to	Notes
Label	rooms, junctions, buildings	Shift+Enter inserts a new line.
Text size	all	8 to 48 px.

Field	Applies to	Notes
Font color	all	Black or white.
Group	rooms	Format: `group_name (Name On Destination)`. Items sharing a group appear as one destination in the route picker.
Disable destination	rooms	Room cannot be chosen as a destination.
Remove display	rooms, stairs	Hides the item on the public map but keeps the node active (see section 11).
Building	nodes	Which building the item belongs to. Only junctions can be set to No building.
Floors	buildings	Number of floors (1 to 20).
Color	all	Fill colour.
Width / Height	all	Size in px. For a selected line, Width shows its length.
Junction X / Y	junctions	Exact position.
Floor	nodes	Limited to the building's floors.
Stair design	stairs	Horizontal or vertical stripes.
Line width	lines	1 to 50 px.
Rotation	all	Degrees.
Multi junction connection	rooms	Allows a room to connect to several junctions.

Below the fields: **Move to front / Move to back** change stacking order, **Apply** commits the fields, and **Delete** removes the selection.

11. Remove display

Tick **Remove display** on a room or stair to hide it on the public map while keeping it fully active. It still takes part in pathfinding and its connections keep working. In the editor the item stays visible so you can still find it, but it is drawn faded with a dashed outline. Untick

the box to show it again. Junctions are never drawn on the public map, so the option is disabled for them.

12. Saving, exporting and publishing

1. **Save layout** keeps a copy in this browser only.
2. **Export JSON** downloads the full layout (nodes, connections, lines, buildings, colours, floor settings and canvas size).
3. Put the exported file on the server as `map.json` (or `map-layout.json`) next to `map.php`. The public map reads it on load.
4. **Import JSON** loads a previously exported file back into the editor.
5. **Reset canvas** clears everything. Use Undo straight away if you pressed it by mistake.

13. How the public map uses your layout

- Rooms and stairs show only when their building is on their floor, and the floor buttons switch buildings.
- Free lines are drawn solid black.
- Junctions with no building follow the floor toggle using the floor you assigned, so a junction on floor 2 only appears while floor 2 is shown.
- Rooms with Disable destination do not appear in the From/To lists. Rooms with Remove display are not drawn but still route.

14. Troubleshooting

Problem	Likely cause
A connection is refused	Check the status pill: room to room, a second junction on a room without Multi junction connection, or floors too far apart.
An item disappeared	It is probably on another floor. Change the Active floor or the building's floor button. Also check that Hide junctions & connections is off.
Floor dropdown only goes up to 2	The item's building has 2 floors. Raise Floors on the building first.

Problem	Likely cause
My edit did not stick	Press Apply or Enter. Edits are discarded if you select another item first.
An item cannot be selected	It may be hidden by Remove display. Look for the faded dashed item, or Ctrl+A to select everything.
Building border options are missing	The Add border and Border width controls were removed. Borders already saved in an old layout still draw.

15. Quick reference

- Draw order: add a building, add rooms, add junctions and stairs, connect them, set floors, export.
- Hold the right mouse button to box-select, middle mouse or Shift/Alt to pan.
- Apply or Enter to commit. Click away to discard.